

Aim

To develop a program to demonstrate the working of **Linear Regression** and **Polynomial Regression** using:

- **Boston Housing Dataset** for predicting house prices
- **Auto MPG Dataset** for predicting vehicle fuel efficiency

Theory

Linear Regression

Linear Regression is a supervised machine learning technique used to model the relationship between independent variables and a dependent variable by fitting a straight line.

- It follows the equation:
$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$
- Used when the relationship between variables is **linear**
- In this experiment, it is used to predict **house prices** based on features like crime rate, number of rooms, etc.

Polynomial Regression

Polynomial Regression is an extension of Linear Regression where the relationship between variables is modeled as a polynomial (curve).

- It follows the equation:
$$Y = b_0 + b_1X + b_2X^2 + \dots + b_nX^n$$
- Used when data shows a **non-linear relationship**
- In this experiment, it is used to predict **fuel efficiency (MPG)** of vehicles

Difference

Linear Regression Polynomial Regression

Straight line	Curved line
Simple model	More complex
Linear data	Non-linear data

Procedure

Part A: Linear Regression (Boston Dataset)

1. Import required libraries
2. Load the Boston Housing dataset
3. Separate features (X) and target (Y)
4. Split dataset into training and testing sets
5. Create Linear Regression model

6. Train the model using training data
7. Predict values using test data
8. Display predicted output

Part B: Polynomial Regression (Auto MPG Dataset)

1. Import required libraries
2. Load Auto MPG dataset
3. Handle missing values (if any)
4. Separate features (X) and target (Y)
5. Convert categorical data (if required)
6. Split dataset into training and testing sets
7. Apply Polynomial Feature transformation (degree = 2)
8. Train Linear Regression model on transformed data
9. Predict values using test data
10. Display predicted output

Result

The program for **Linear Regression** using the Boston Housing Dataset was executed successfully, and the model predicted the house prices based on input features.

The program for **Polynomial Regression** using the Auto MPG Dataset was also executed successfully, and the model predicted the fuel efficiency (miles per gallon) of vehicles.

The outputs obtained show that:

- Linear Regression is suitable for linear relationships.
- Polynomial Regression provides better results for non-linear data.

Thus, both regression techniques were implemented and verified successfully.
